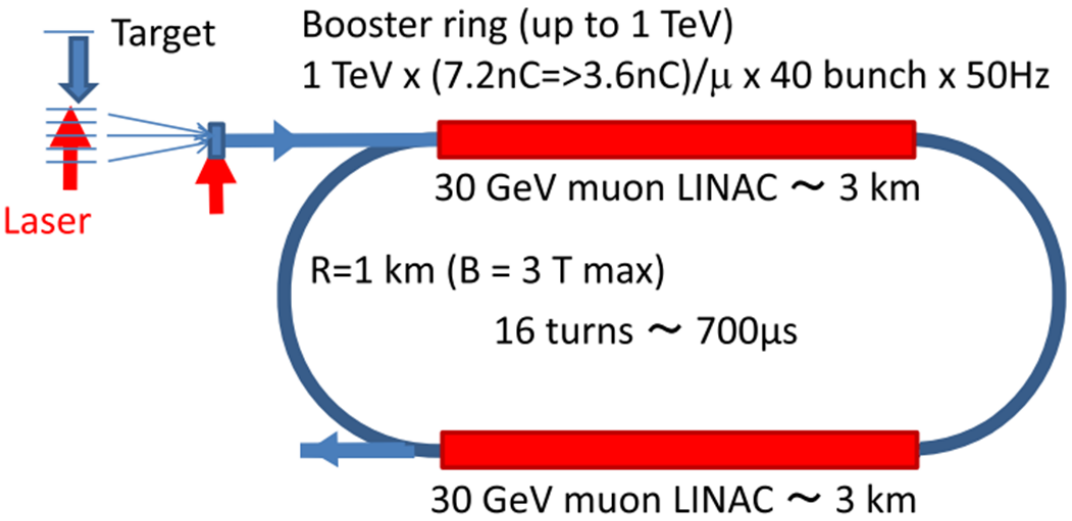
Bending Arcs of Antimuon Boosting Ring

Burak Dağlı
02/12/2025

Bending Magnet Concept in the μTRISTAN study



2.3 *Muon accelerator*

The ultra-cold muons are pre-accelerated and injected into the booster ring of race-track shape (see Fig. 1). The length of the LINAC part is 3 km and there are arc sections with $R = 1$ km where an alternative gradient bending magnet of 3 T is used. The booster ring is used to accelerate the μ^+ beam up to the energy of 1 TeV. In each of the two LINAC parts of the booster ring the μ^+ beam is accelerated by 30 GeV, which with the 1 TeV beam is achieved by 16 turns. This takes about 700 μs. The LINAC is also intermittently used for electron acceleration for the μ^+e^- collider.

During the acceleration, about half of the muons decay and the bunch charge is reduced to about 3.6 nC. Taking into account the beam loss, the total beam loading is about 9 MW.

The **Alternative Gradient Bending Magnet** is a **combined-function magnet** utilized in modern high-energy synchrotrons to provide both the necessary **dipole field for steering** and the **Alternating Gradient strong focusing** for beam stability in a single component.

Fixed-Field Alternative Gradient (FFAG) ???

Hamada, Y., Kitano, R., Matsudo, R., Takaura, H., & Yoshida, M. (2022). μTRISTAN. *Progress of Theoretical and Experimental Physics*, *2022*(5), 053B02. https://doi.org/10.1093/ptep/ptac059

Scaling vs Non-Scaling FFAG

Feature	Scaling FFAG (SF-FFAG)	Non-scaling FFAG (NS-FFAG)
Magnetic Field Change Over Time	**Fixed** (Constant in Time)	**Fixed** (Constant in Time)
Orbit Radius (vs. Energy)	**Increases significantly** with energy.	**Minimal increase** with energy (Compressed orbit).
Magnetic Field Formula	**$B(r) \propto r^k$** (Satisfies the scaling condition)	**Simplified field form** (Does not satisfy the scaling condition)
Magnet Aperture	**Wide** (Large space required)	**Narrow** (Small space required)
Betatron Tune	**Constant** across the energy range. No resonance crossing.	**Varies** with energy. Resonance lines are crossed (High risk).
High Energy Limit	**Practically Limited** Generally up to 100 - 300 MeV (Due to massive magnet size and cost).	**Suitable for High Energy** Potential to reach **GeV** levels (Due to smaller magnets and cost effectiveness).
Acceleration Type	**Fast acceleration** is possible (Synchro-cyclotron-like)	**Fast acceleration** is mandatory (To cross resonances quickly)

Sheehy, D. S. (2014). FFAG Accelerators, CERN Introductory Accelerator School. CAS, Prague.

Altogether, about a dozen scaling and a dozen non-scaling designs are currently under study for the acceleration of protons, heavy ions, electrons, and muons, with top energies ranging from **1 MeV to 20 GeV**, and diameters from 5 cm to 400m.

Craddock, M. K., & Symon, K. R. (2008). Cyclotrons and fixed-field alternating-gradient accelerators. Reviews of accelerator science and technology, 1(01), 65-97.

We need to reach hundreds of GeV energies !!

Limitation of Non-Scaling FFAG

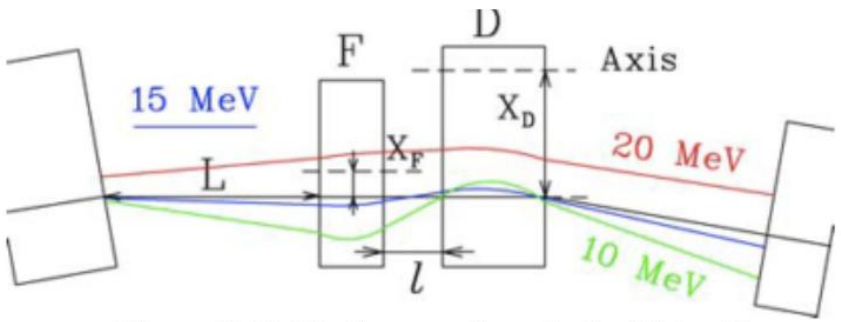


Figure 2: Orbits in a quadrupole doublet cell.

Sheehy, D. S. (2014). FFAG Accelerators, CERN Introductory Accelerator School. CAS, Prague.

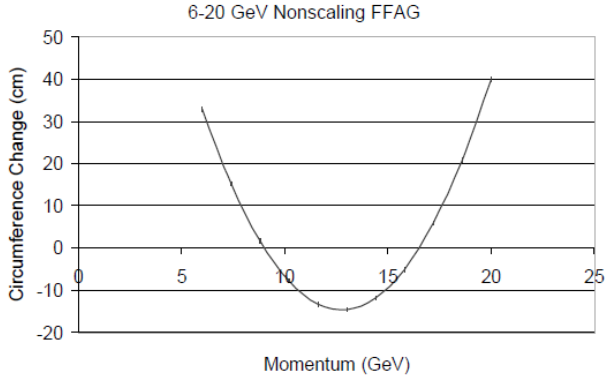
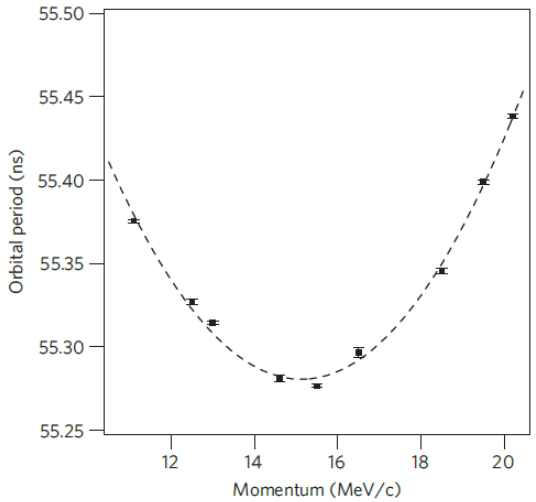


Fig. 5. Circumference change as a function of momentum.

Johnstone, C., & Koscielniak, S. (2003). FFAGs for rapid acceleration. Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, 503(3), 445-457.

Acceleration in the linear non-scaling fixed-field alternating-gradient accelerator EMMA



Machida, S. et al., Acceleration in the linear non-scaling fixed field alternating gradient accelerator EMMA, Electron Model for Many Applications.

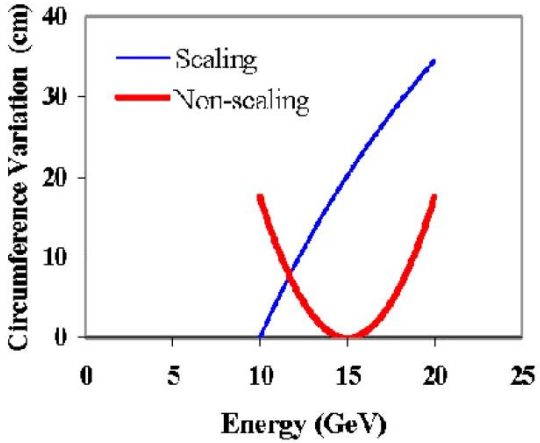


Fig. 37. Circumference variation with energy compared for scaling and linear non-scaling FFAGs.

Craddock, M. K., & Symon, K. R. (2008). Cyclotrons and fixed-field alternating-gradient accelerators. Reviews of accelerator science and technology, 1(01), 65-97.

Muon-Ion Collider at BNL

Table 2. Proposed beam parameters and estimates of achievable luminosity for MuIC at BNL and LHmuC at CERN for muon-proton collisions.

Parameter	MuIC				LHmuC	
$\sqrt{s_{\mu p}}$ (TeV)	0.33	0.74	1.0		6.5	
$L_{\mu p}$ (10^{33}cm^{-2}s^{-1})	0.07	2.1	4.7		2.8	
Int. Lumi. (fb^{-1}) per 10 yrs	6	178	400		237	
		Muon		Proton	Muon	Proton
Beam energy (TeV)	0.1	0.5	0.96	0.275	1.5	7
N_b (10^{11})	40	20	20	3	20	2.2
f^{μ}_{rep} (Hz)	15	15	15		12	
Cycles per μ bunch, N^{μ}_{cycle}	1134	1719	3300		3300	
$\varepsilon^{*}_{x,y}$ (μm)	200	25	25	0.3	25	2.5
$\beta^{*}_{x,y}$ @IP (cm)	1.7	1	0.75	5	0.5	15
Trans. beam size, $\sigma_{x,y}$ (μm)	48	7.6	4.7	7.1	3	7.1

colliders of 2–4 MW in beam power and 4–8 GeV in kinetic energy yet. A substantial upgrade is necessary. A dedicated muon front-end is needed to provide the muon injection, cooling, and initial fast acceleration with recirculating linacs. Ideally, both muon acceleration and storage rings would fit within the existing tunnel to minimize the infrastructure cost. There are six straight sections of the BNL tunnel that could be used for the accelerator. On the other hand, the design by the MAP collaboration favors an acceleration ring with larger curvature for fast-ramping magnets and long straight sections for fast acceleration, and a compact circular ring for muon storage and collisions. Therefore, it is conceivable that a new tunnel would be added for muon acceleration, as

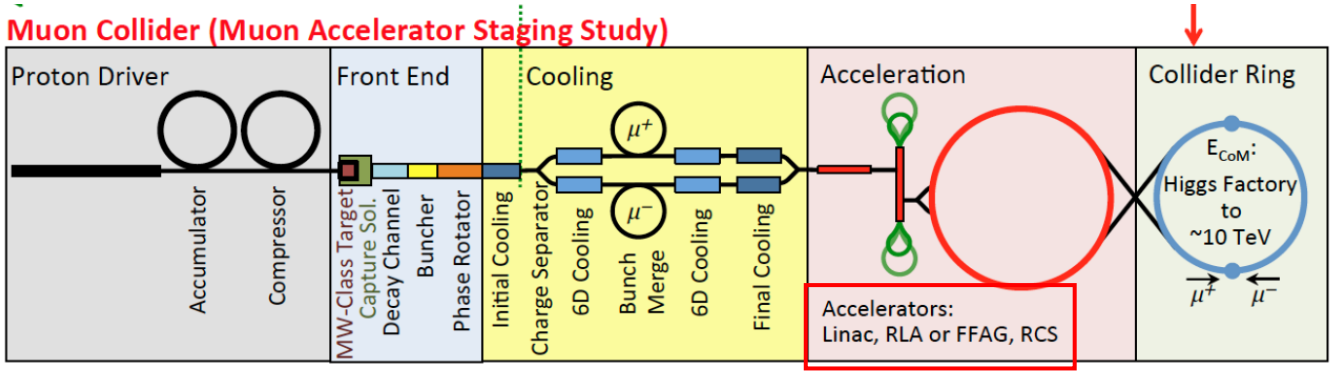
remains challenging to achieve a high energy, high-luminosity muon collider because of the short muon lifetime. The efficient muon cooling in six dimensions (spatial and momentum) and the fast ramping of the muon energy are among the most critical elements for its realization, necessitating an R&D program in the U.S and at CERN through the Muon Accelerator Program (MAP) and the international muon collider collaboration (IMCC), respectively.

This proposal enables deep inelastic scattering measurements in new regimes at low parton

No extra details given for the fast-ramping magnets!!

D. Acosta et al The potential of a TeV-scale muon-ion collider 2023 JINST 18 P09025

Muon Collider Forum



The MAP feasibility assessment demonstrated sufficient progress in several key areas to suggest that muon colliders are a viable technology for a high-energy collider. Several issues were identified as part of the MAP Feasibility Assessment that had the greatest risks for the realization of a muon collider concept. These included: operation of RF cavities in high magnetic fields in the front end and cooling channel; development of a 6D cooling lattice design consistent with realistic magnet, absorber, and RF cavity specifications; a direct demonstration and measurement of the ionization-cooling process; development of very-high-field solenoids to achieve the emittance goals of the Final Cooling system; and demonstration of fast-ramping magnets to enable RCS capability for acceleration to the TeV scale.

beam. Several cooling stages then reduce the longitudinal and transverse emittance of the beam using a sequence of absorbers and RF cavities in a high magnetic field. A linac and two recirculating linacs accelerate the beams to 60 GeV. One or more rings accelerate the beams to the final energy. As the beam is accelerated, the lifetime in the lab frame increases due to relativistic time dilation so later stage accelerators have proportionally more time for acceleration, so that rapid-cycling or fast pulsed synchrotrons can be used. Fixed-Field Alternating-gradient (FFA) accelerators are an interesting alternative. Finally the two single bunch beams are injected at full energy into the collider

requirements of high magnetic fields.

Reaching higher energies would ideally be done via pulsed synchrotrons. Rapid cycling synchrotrons are extensively utilized in existing accelerators, but the requirements for acceleration in muon colliders differ in two important aspects: short (ms-scale) acceleration cycles with a relatively low repetition rate (around 10 Hz), thus "pulsed" rather than "rapid cycling"; and the requirement for a high average bend field. To achieve a high average bend field, fixed field superconducting magnets are interleaved with bipolar pulsed iron dipoles, denoted a "hybrid" pulsed synchrotron. Pulsed iron magnets can achieve fields of 1.5 T with manageable power consumption [195]; they could reach

Black, K. et al. (2024). Muon collider forum report. Journal of Instrumentation, 19(02), T02015.

Muon Acceleration at RCS rings of the MC

F Appendix: High Energy Acceleration

As described in [25], an option for the chain of four rapid cycling synchrotrons (RCS) foresees to accelerate two counter-rotating bunches at a repetition rate of 5 Hz in stages of 0.30 TeV (RCS1), 0.75 TeV (RCS2) and 1.5 TeV (RCS3) to inject into the 3 TeV collider ring, or 5 TeV (RCS4), to inject to the 10 TeV collider ring. This scenario is based on the US Muon Acceleration Program (MAP) [26, 27] and applied for a general Greenfield site. The high-energy stage of the accelerator chain with four RCS is illustrated in Fig. F.1. Corresponding site-specific parameter designs can be found in Section F.1. Alternative Fixed Field Accelerator options can be found in Section F.2.

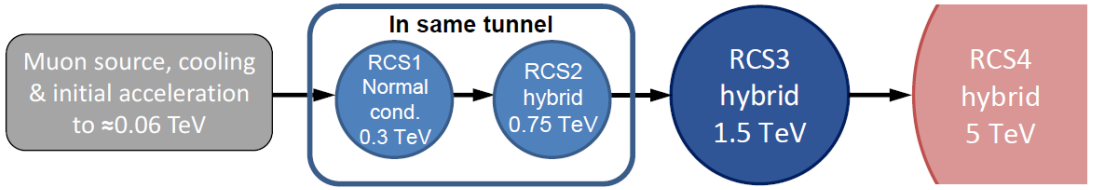


Fig. F.1: Schematic of the chain of rapid cycling-synchrotrons for the high-energy acceleration complex. From [25].

The first two RCS share the same tunnel, meaning that they have the same circumference and layout [28]. The bending in the first RCS is provided by normal conducting magnets. The RCS2 to RCS4 are planned as hybrid RCSs where normal conducting magnets cycling from $-B_{nc}$ to $+B_{nc}$ are interleaved with strong fixed-field, superconducting magnets. Within this section, NC magnets are referred to as *pulsed*, and the SC magnets are referred to as *steady*. This is to reflect the alternative magnet technologies required for the hybrid RCS. This combination allows for a large energy swing with a high average

Parameter	Unit	RCS1	RCS2	RCS3	RCS4
Hybrid RCS	-	no	yes	yes	yes
Repetition rate	Hz	5	5	5	5
Circumference	m	5990	5990	10700	35000
Injection energy	GeV	63	314	750	1500
Extraction energy	GeV	314	750	1500	5000
Energy ratio		5.0	2.4	2.0	3.33
Assumed survival rate		0.9	0.9	0.9	0.9
Cumulative survival rate		0.9	0.81	0.73	0.66
Acceleration time	ms	0.34	1.10	2.37	6.37
Revolution period	µs	20	20	36	117
Number of turns	-	17	55	66	55
Required energy gain/turn	GeV	14.8	7.9	11.4	63.6
Average accel. gradient	MV/m	2.44	1.33	1.06	1.83
Number of bunches		1	1	1	1
Inj. bunch population	10^{12}	2.7	2.4	2.2	2
Ext. bunch population	10^{12}	2.4	2.2	2	1.8
Beam current per bunch	mA	21.67	19.5	9.88	2.75
Peak RF power	MW	640	310	225	350
Vert. norm. emittance	µm	25	25	25	25
Horiz. norm. emittance	µm	25	25	25	25
Long. norm. emittance	eVs	0.025	0.025	0.025	0.025
Bunch length at injection	ps	31	22	18	14
Bunch length at ejection	ps	22	18	14	10
Straight section length	m	1012.4	536.4	793.6	4385.6
Length with pulsed dipole magnets	m	3654	2539	4366	20376
Length with steady dipole magnets	m	-	1115	2358	4257
Max. pulsed dipole field	T	1.8	1.8	1.8	1.8
Max. steady dipole field	T	-	10	10	16
Ramp rate	T/s	4232	3272	1519	565
Main RF frequency	GHz	1.3	1.3	1.3	1.3
Harmonic number		25900	25900	46300	151400

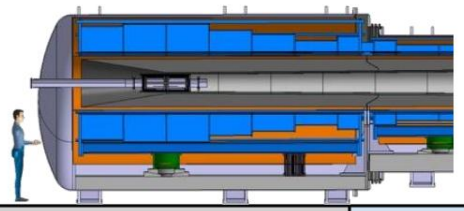
Table 5.1: RCS acceleration chain key parameters

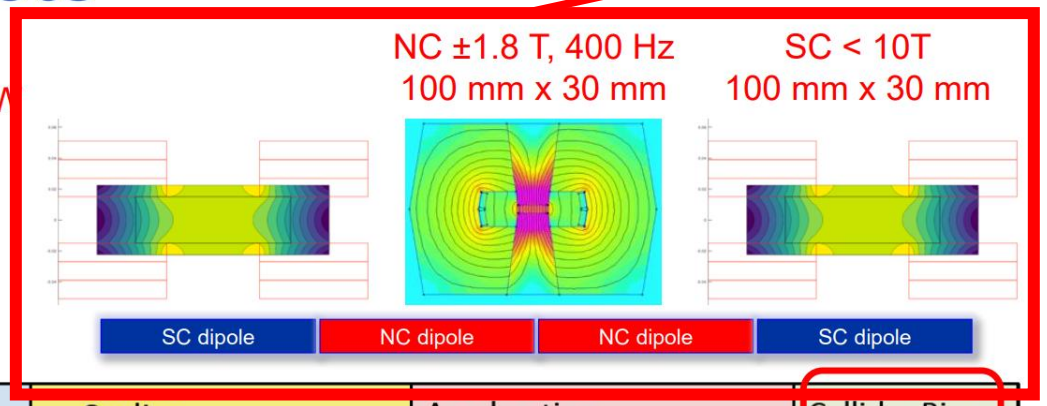
MuCol (2025). MuCol Milestone Report No. 7: Consolidated Parameters. arXiv preprint arXiv:2510.27437.

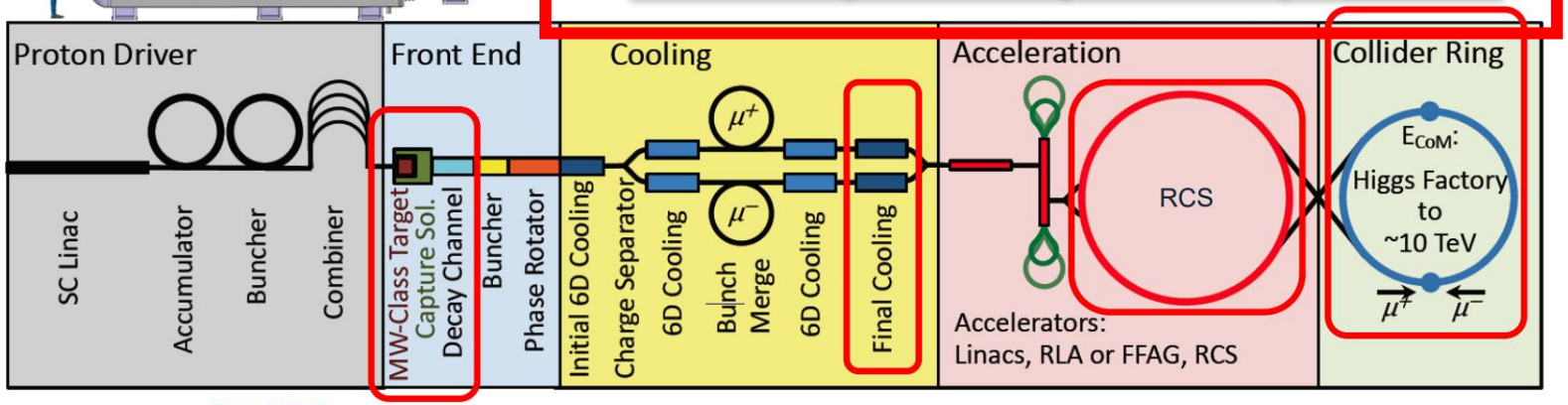
Pulsed NC + Steady SC → Hybrid Bending Magnet ??

Muon Collider magnets

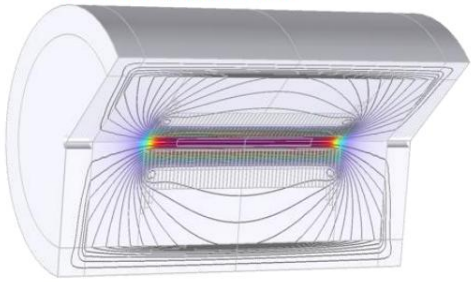
20 T, 200 mm
Radiation heat load ≈ 5…10 kW
Radiation dose: 80 MGy



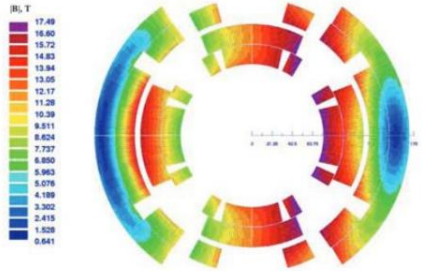




> 40 T, 60 mm



16 T peak, 160 mm
Radiation heat load ≈ 5 W/m
Radiation dose ≈ 20…40 MGy



Luca Bottura

Mike Lamont, Welcome and Introduction, Workshop on ultra-high-field solenoids 25–26 Kas 2025 CERN

Why UHF solenoids at CERN?

MuCol is still heavily focused on effective and rapid Muon Cooling!!

- The performance of the muon collider depends critically on achieving **high- and ultra-high-field levels**:
  - 20 T at the beam target, 1.4 m bore;
  - 40 T for final cooling, 6 cm bore.

- Solenoids are an **ideal R&D vehicle** for novel technology, e.g. HTS, also for accelerator magnets:
  - Field reach, force and stored energy for a given material cost is advantageous when compared to dipoles, turnover can be fast.
- There are **clear and strong synergies** between work for HEP and needs of other fields of scientific and societal applications.

This Workshop on UHF Solenoids is an important milestone

Hybrid Magnets of the MC

Contrary to other RCS, the RCS1 uses only pulsed magnets. The other RCSs are hybrid ones. That means that pulsed magnets (generally normal conducting, NC) are interleaved with static magnets (generally superconducting, SC). The main motivation is to combine a rapid cycling (given by the pulsed dipoles) with a small circumference (given by the strong static dipoles). The possibility to reuse the LHC tunnel to host a fourth RCS is under discussion. This fourth

though the scaling laws have been generalized to an arbitrary number of families. We will respectively use the subscripts 1 and 2 for the dipole families and note D_1 and D_2 the two dipoles. The dipole sequence is done by alternating the two families: $D_1 - D_2 - \ldots - D_2 - D_1$. Each half-cell has thus $n_d + 1$ dipoles D_1 and n_d dipoles D_2. The total number of

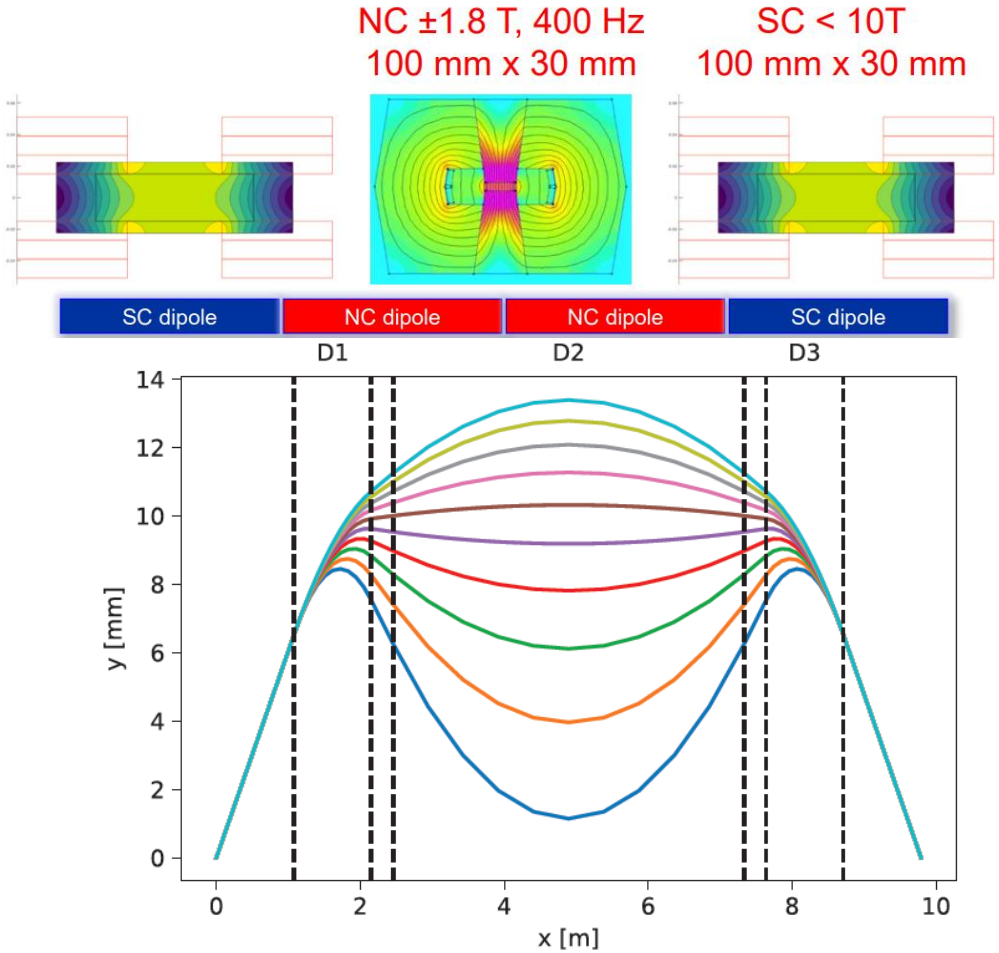


Figure 2: Trajectories in a half-cell for RCS2 with the parameters given in Table 1. The injection/extraction orbits are respectively in the inner/outer side (in blue/cyan).

A. Chance et al., "Parameters range for a chain of rapid cycling synchrotrons for a muon collider complex," in Proc. 14th Int. Part. Accel. Conf., 2023, pp. 913–916, doi: 10.18429/JACoW-IPAC2023-MOPL162.

Example studies of Pulsed Bending Magnets

Table 1: Specifications for the pulsed bending magnet.
() value is for the second magnet.

Max. beam energy	3 GeV
Deflection angle	114.4 mrad
Field strength @ 3 GeV	1.15 T
Magnet core length	990 (980) mm
Magnet core gap & width	30 x 155 mm
Coil turn	1
Max. PS peak current	32 kA
Peak current @ 3 GeV	28 kA
Pulse width	200 μsec
Pulse shape	half sine
Max. repetition	25 Hz
Effective current	1.4 kA
Stability	1×10^{-3} (p-p)
Peak force for coil	1.6 ton
Magnet resistance	1 mΩ
Magnet inductance	6.6 μH

Tawada, M., Kikuchi, M., Mimashi, T., Nagahashi, S., & Ueda, A. (2009). Development of pulsed bending magnet for simultaneous top-up injection to KEKB and PF ring. Proc. PAC09, Vancouver, BC, Canada, 175.

TABLE 3. Dipole magnet specification

Total required dipole length	2200 m	Beam repetition rate	15 Hz
Gap	25 mm	Good field region (h x v)	60×10 mm^2
Design field B_{design}	1.5 T	Maximum field B_{max}	1.75 T
Field quality at B_{max}	1×10^{-3}	Ramp rate (equivalent frequency)	1000 Hz
Power Loss Yoke (at B_{design})	2.7 MW	Power Loss Coil (at B_{design})	0.55 MW
Total Power Loss (at B_{design})	3.25 MW	Stored energy (at B_{design})	4200 J/m
Current per bus bar	15600 A turns	Number of bus bars	4
Average peak current density cable	16 A/mm^2	DC resistance bus bar	1.77×10^{-5} Ω/m
Voltage required to drive current	866 V/m	Inductance	8.84 μH

Berg, J. S., & Witte, H. (2016, October). Pulsed synchrotrons for very rapid acceleration. In AIP Conference Proceedings (Vol. 1777, No. 1, p. 100002). AIP Publishing LLC.

Is it enough?
(It seem like it isn't yet. We must keep looking)

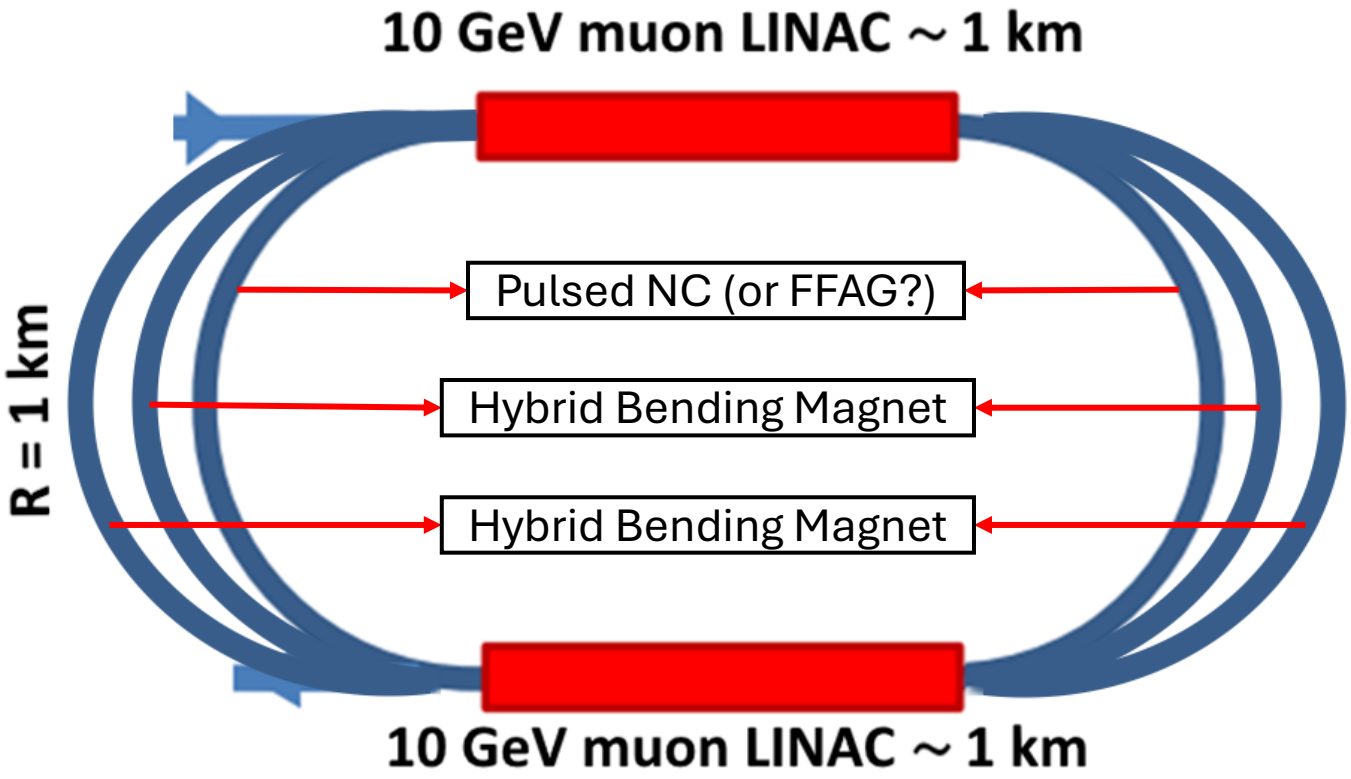
Maybe we should also work on:

Number of bunch ↓
Repetition rate ↓
Muons per bunch ↑

μTRISTAN→10^{10} MC→10^{12}

What can be done to reach 1 TeV energy?

Pulsed NC + Steady SC → Hybrid Bending Magnet



RCS1 and RCS2 have same arc length with LHeC tunnel

Parameter	Unit	RCS1	RCS2	RCS3	RCS4
Hybrid RCS	-	no	yes	yes	yes
Repetition rate	Hz	5	5	5	5
Circumference	m	6283	6283	15500	35437
Injection energy	GeV	63.0	174	454	1541
Extraction energy	GeV	174	454	1541	5000

MuCol (2025). MuCol Milestone Report No. 7: Consolidated Parameters. arXiv preprint arXiv:2510.27437.

vFFA as alternative for RCS rings of the MC

F.2 Fixed Field Acceleration

Milestone 17 [30] outlines the possibility of using vertical-excursion Fixed-Field Accelerator (vFFA) [31] rings as alternatives to one or more of the RCS rings, giving example parameters for RCS1 and RCS4 greenfield equivalents. These would provide the possibility for acceleration unconstrained by magnet ramp rates, removing issues for power conversion and storage, and enabling the construction of rings with full-superconducting magnet technology (thereby enhancing power efficiency). The relative isochronicity of the vFFA concept mitigates the need for frequency cycling in the RF systems, and enables the possibility of on-crest acceleration for an increased acceleration efficiency at a given RF voltage. Table F.5 lists the design parameters of these FFA rings alongside key parameters for comparison to the RCS equivalents.

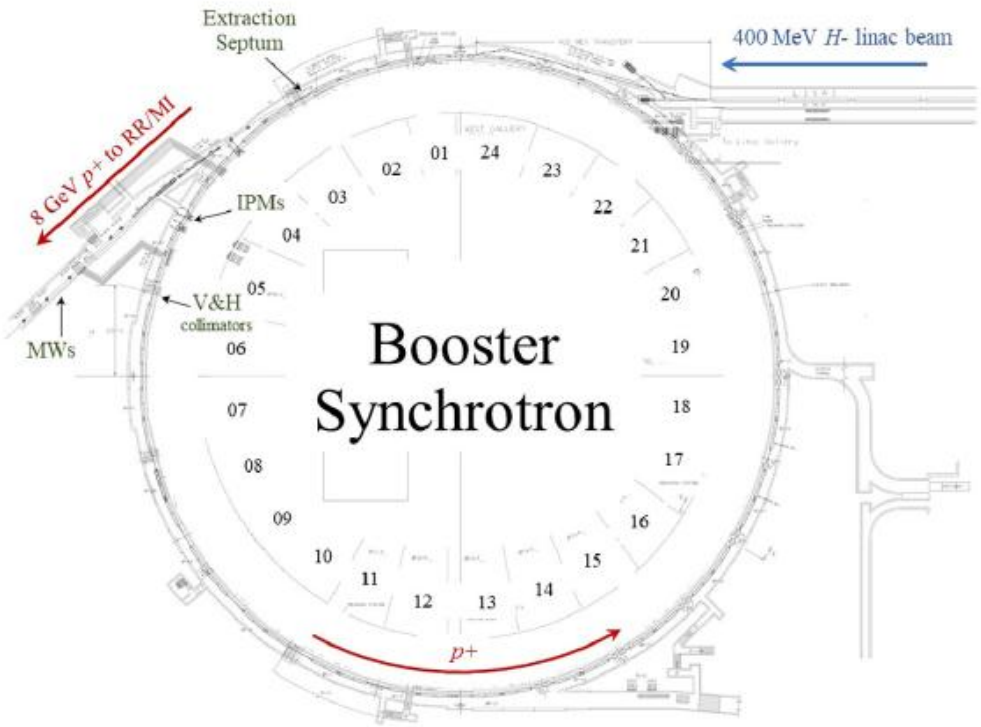
However, the use of FFA arcs implies a closed orbit that moves as a function of energy. This increases requirements for element apertures. Milestone 17 presents a scheme for the implementation of dispersion suppressors to reduce the impact of this upon the RF systems. Depending on the specific execution of these schemes, further optimisation of the parameters in the included table could be possible to reduce peak fields and reduce the size of the machine by separating RF requirements from arc design requirements.

Table F.5: FFA alternative tentative parameters

Design Parameter	Symbol	Unit	vFFA1	vFFA4
Orbit radius at centre of F-magnet	r_0	m	953	5570
F-magnet bending angle	θ_F	rad.	0.01033	0.00786
Number of cells	N_c		790	1000
F-magnet half-opening angle	β_F	rad.	0.0015640	0.0019700
D-magnet half-opening angle	β_D	rad.	0.0011800	0.0011660
F-magnet orbit inclination	γ_F	rad.	0.252	-0.492
vFFA normalised field index	m	1/m	31.94	12.13
Comparison Parameter				
Circumference [m]		m	5990	35000
Injection Energy [TeV]		TeV	0.06	1.5
Extraction Energy [TeV]		TeV	0.3	5
Ramp Rate [T/s]		T/s	0	0
Vertical Excursion [m]		m	0.048	0.099
Relative path length difference			0	0
Peak Dipole Field On Orbit [T]		T	6.93	13.59
Peak Dipole Field (Good Field Region) [T]		T	12.52	29.04
Drift length [m]		m	1.18	1.03
Tune			(0.382, 0.079)	(0.460, 0.057)

MuCol (2025). MuCol Milestone Report No. 7: Consolidated Parameters. arXiv preprint arXiv:2510.27437.

Fermilab booster: rapid-cycling synchrotron



the existing 8 GeV Booster synchrotron [22,23].

The Fermilab Booster [24] is a 474.2 m circumference, alternating-gradient, rapid-cycling synchrotron containing 96 combined-function magnets; see Fig. 1. Together with capacitor banks, these magnets form a resonant network

TABLE I. Main operational parameters of the Fermilab Booster. β_p and γ_p are relativistic Lorentz factors of protons.

Parameter		Comments
Circumference, C	474.20 m	
Injection energy (kinetic), E_i	400 MeV	$\beta_p = 0.701$, $\gamma_p = 1.426$
Extraction energy (kinetic), E_f	8 GeV	$\beta_p = 0.994$, $\gamma_p = 9.526$
Cycle time, $T_0 = 1/f_0$	1/15 s	20,000 turns
Harmonic number	84	
rf frequency, f_{rf}	37.77–52.81 MHz	injection-extraction
Maximum rf voltage, V_{rf}	1.1 MV	
Transition energy, E_{tr}	4.2 GeV	$\gamma_{tr} = 5.478$, at $t = 17$ ms
No. of cells, magnets	24, 96	$FOFDOOD$, 96°/cell
Total intensity, N_p	4.5×10^{12}	$N_b = 81$ bunches
rms normalized emittance, $\varepsilon_{x,y}$	2.0π μm	12π μm for 95%
β functions, $\beta_{x,y}$	33.7/20.5 m	maximum
Dispersion function, D_x	3.2 m	maximum

Eldred, J., Lebedev, V., Seiya, K., & Shiltsev, V. (2021). Beam intensity effects in Fermilab Booster synchrotron. Physical Review Accelerators and Beams, 24(4), 044001.

Design Report:
https://beamdocs.fnal.gov/AD/DocDB/0010/001022/001/Booster%20V3_0.pdf